

Program 1 – Bar Plot

Aim: To create a bar plot for net profit analysis of companies TCS, Infosys, HCL, Wipro, Tech Mahindra in 2025 with unique color for each bar.

Code:

```
profit <- c(5.8, 3.8, 1.9, 1.4, 0.8)
companies <- c("TCS", "Infosys", "HCL Tech", "Wipro", "Tech Mahindra")
barplot(
  profit,
  names.arg = companies,
  col = c("blue", "green", "orange", "purple", "red"),
  xlab = "Companies",
  ylab = "Net Profit (Billion USD)",
  main = "Net Profit Analysis 2025"
)
```

Program 2 – Pie Chart

Aim: To construct a pie diagram for the following tax revenue data of India private with the year 2021.

Code:

```
tax <- c(6526, 7108, 2568, 560, 763)
names <- c("Excise", "Customs", "Corporation Tax", "Income Tax", "Others")
pie(
  tax,
  labels = names,
  main = "TAX Collection PIE Chart",
  col = rainbow(length(tax))
)
```

Program 3 – Histogram

Aim: To construct a histogram for the following data of daily sales of certain shops.

Code:

```
midx <- seq(12.5, 112.5, 25)
fn <- c(5, 8, 13, 11, 3)
y <- rep(midx, fn)
brk <- seq(0, 125, 25)
hist(
  y,
  breaks = brk,
  xlab = "Sales",
  ylab = "Frequency",
  main = "Sales Analysis",
  col = "Blue"
)
```

Program 4 – Scatter Diagrams

Aim: To create scatter diagrams for three datasets with different plot characters and colors.

Code:

```
x1 <- c(10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5)
y1 <- c(8.04, 6.95, 7.58, 8.81, 8.33, 9.96, 7.24, 4.26, 10.84, 4.82, 5.68)
plot(x1, y1, pch = 16, main = "Scatter Diagram 1", xlab = "x1", ylab = "y1")

x2 <- c(10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5)
y2 <- c(9.14, 8.14, 8.74, 8.77, 9.26, 8.10, 6.13, 3.10, 9.13, 7.26, 4.78)
plot(x2, y2, pch = 17, main = "Scatter Diagram 2", xlab = "x2", ylab = "y2", col = "red")

x3 <- c(10, 5, 3, 2, 9, 8, 6, 11, 13, 4, 12)
y3 <- c(6.13, 7.95, 3.12, 5.74, 8.0, 4.26, 2.12, 6.24, 5.24, 4.82, 8.31)
plot(x3, y3, pch = 18, main = "Scatter Diagram 3", xlab = "x3", ylab = "y3", col = "green")
```

Program 5 – Ogive (Less Than Cumulative Frequency)

Aim: To construct a less than cumulative frequency curve for the following age distribution data.

Code:

```
f <- c(5, 12, 15, 20, 10, 8)
lcf <- cumsum(f)
ul <- seq(10, 60, 10)
plot(
  ul, lcf,
  type = "b",
  xlim = c(10, 60),
  xlab = "Age",
  ylab = "No. of People Below",
  main = "Age Distribution Ogive",
  lwd = 3
)
```

Program 6 – Descriptive Statistics

Aim: To compute the mean, median, quartile deviation and SD of the marks of students: 12, 15, 5, 9, 10, 25, 33, 21.

Code:

```
marks <- c(12, 15, 5, 9, 10, 25, 33, 21)
mark_mean <- mean(marks)
mark_median <- median(marks)
mark_variance <- var(marks)
mark_sd <- mark_variance^0.5
q1 <- quantile(marks, 0.25)
q3 <- quantile(marks, 0.75)
mark_qd <- (q3 - q1) / 2
print(paste("Mean:", mark_mean))
print(paste("Median:", mark_median))
print(paste("SD:", mark_sd))
print(paste("QD:", mark_qd))
```

Program 7 – Correlation & Regression

Aim: To obtain the Karl Pearson's correlation coefficient and Spearman's correlation coefficient, obtain y on x and x on y regression lines, plot the y on x regression line, and predict marks in mathematics if marks in statistics is 50.

Code:

```
x <- c(46, 42, 44, 40, 43, 41, 45)
y <- c(40, 38, 36, 35, 39, 37, 41)
kn <- cor(x, y, method = "pearson")
sn <- cor(x, y, method = "spearman")
fit_yonx <- lm(y ~ x)
fit_xony <- lm(x ~ y)
plot(x, y, col = "red")
abline(fit_yonx, col = "black", lwd = 3)
x_exp <- (0.75 * 50) + 14.50
print(x_exp)
```

Program 8 – Normal Distribution

Aim: a) Draw a random sample of size 250 from a Normal distribution with mean 50 and SD 10. Plot the density graph. b) Find probabilities for weight: i) > 60kg, ii) < 40kg, iii) between 50kg and 62kg.

Code:

```
x <- rnorm(250, 50, 10)
plot(density(x))

# Probabilities (mean = 55, sd = 7)
p1 <- pnorm(60, 55, 7)
a <- 1 - p1
a

b <- pnorm(40, 55, 7)
b

p2 <- pnorm(62, 55, 7)
p3 <- pnorm(50, 55, 7)
c <- p2 - p3
c
```